INCH-POUND

MIL-T-48149A (AR)
AMENDMENT 4
5 October 1989
SUPERSEDING
AMENDMENT 3
15 January 1989

MILITARY SPECIFICATION

TITANIUM DIOXIDE, DRY

This amendment forms a part of Military Specification MIL-T-48149A (AR), dated 14 January 1976, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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2.1 Standards.

Delete "FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing."

Delete "MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing."

Add "MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing."

3.3 Specific gravity. Change "The specific gravity shall be between 3.8 and 4.2" to "The specific gravity shall be between 3.8 and 4.3."

* 3.4 Oil Absorption. First sentence: Change "The oil absorption shall be between 20.0 and 40.0 pounds ....." to "The oil absorption shall be between 10.0 and 40.0 pounds ....."

3.5 Water solubility. Delete in its entirety.

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4.4.1 Inspection lot formation. Delete in its entirety and substitute:

"4.4.1: Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. A unit is a batch. For continuously produced titanium dioxide a unit (production interval) is one (1) hour of continuous production."

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4.4.3 Testing. Delete in its entirety and substitute:

"4.4.3 Testing. The product shall be submitted for inspection to determine compliance with all requirements in 3. Failure of a sample to comply with any of the requirements shall be considered a major defect and shall result in rejection of the batch/lot.

4.4.3.1 Continuous process sampling. A representative sample, of approximately one pound shall be selected in accordance with ASTM E300 for solids, every hour. The inspection of the samples shall be in accordance with MIL-STD-1235, CSP-1 Plan using an AQL off 6.5 percent. If any sample fails to meet any test requirement the batch (see 4.4.1) represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the acceptable methods given in 4.5. If any of these batches fail to meet any test requirement, that batch shall also be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235.

4.4.3.2 Batch process sampling. For titanium dioxide produced by the batch process, a representative sample of approximately one pound shall be selected in accordance with ASTM E300 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan using an AQL of 6.5 percent. If any sample fails to meet any requirement, the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.5. If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235.

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4.4.3.3 Product testing. The samples obtained in 4.4.3.1 or 4.4.3.2 shall be subjected to the tests listed in Table 1 unless otherwise specified.

TABLE 1

Inspection/Requirement	Defect Classification	Test Paragraph
TiO_2 Content (see 3.1)	Major	4.5.1
Coarse particles (see 3.2)	Major	4.5.2
Specific gravity (see 3.3)	Major	4.5.3
Oil absorption (see 3.4)	Major	4.5.4
Hydrogen ion concentration (see 3.6)	Major	4.5.6
Moisture (see 3.7)	Major	4.5.7
Compatibility (see 3.8)	Major	4.5.8

4.5 After "(see 6.5)", add:

"The tests in 4.5.1 through 4.5.8 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks."

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Paragraph 4.5.5 Water solubility. Delete in its entirety:

Paragraph 4.5.8 Compatibility. Delete in its entirety and substitute:

"4.5.8 Compatibility (see 6.7). Compatibility shall be determined in accordance with MIL-STD-286, Method T408.1 at $90^{\circ} \pm 0.5^{\circ}C$ using M30 propellant, with the following exception. Rasp or cut the propellant and then pass it through number 12 and 20 sieves. The material retained on the number 20 sieve shall be mixed to form a homogeneous sample to be used in this test."

6.2 Delete in its entirety and substitute:

"6.2 Sources. Sources of titanium dioxide which have been found satisfactory for intended use are:

* 1) Ti-Pure, Rutile Titanium Dioxide, R-100, Dupont, Wilmington, DE.

2) Unitane 0-220, Kemira, Inc., Savannah, GA

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3) Tiatanox 2005, National Lead, South Amboy, NJ.

4) Tronox CR833, Kerr-McGee, Oklahoma City, OK.

Use of any other materials other than the four listed above shall require prior approval by Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-Q, Picatinny Arsenal, NJ 07806-5000".

Paragraph 6. Add new paragraph as follows:

"6.7 Compatability testing. The Contracting Officer may recommend that the compatibility test be performed at the U.S. Army Armament Research, Development and Engineering Center. Samples for this test shall be submitted in accordance with instructions of the Contracting Officer to Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: AMSMC-QAR-Q, Picatinny Arsenal, NJ 07806-5000."

The margins of this amendment are marked with an asterisk to indicate where change (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

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